

A
Project synopsis on
AI.VIRTUAL-MOUSE

**Bachelor of Technology Computer Engineering
with specialization in data science**

by

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Project Title: A.I -Virtual- Mouse

1. Introduction:

The AI Virtual Mouse project aims to develop a software solution that allows users to control their computer mouse cursor using hand gestures and facial recognition. This innovative approach to human-computer interaction leverages artificial intelligence (AI) to provide a more intuitive and accessible way of interacting with computers.

2. Objectives:

- Create a virtual mouse system that eliminates the need for traditional physical mice.
- Enable gesture-based cursor control using a standard computer webcam.
- Implement facial recognition for user identification and customization.
- Improve accessibility for individuals with physical disabilities.
- Enhance user experience and efficiency by reducing reliance on physical input devices.

3. Key Features:

a. Gesture-Based Control:

- The system will track and interpret hand gestures to control the cursor's movement.
- Recognize common gestures such as pointing, clicking, and dragging.
- Support customizable gestures for user preference.

b. Facial Recognition:

- Use facial recognition to identify and authenticate users.
- Allow users to configure settings and preferences tied to their facial profiles.
- Switch between user profiles seamlessly based on facial recognition.

c. Accessibility:

- Target users with mobility impairments who may struggle with traditional mice.
- Implement accessibility features like voice commands and on-screen keyboard integration.

d. Calibration and Customization:

- Enable users to calibrate and fine-tune gesture and facial recognition settings.

- Store user-specific preferences for a personalized experience.

e. Compatibility:

- Develop the AI Virtual Mouse to be compatible with popular operating systems (Windows).
- Ensure it works with standard webcams, and provide options for higher-end cameras with improved accuracy.

6. Conclusion:

The AI Virtual Mouse project introduces a novel way of interacting with computers, improving accessibility and user experience. By utilizing gesture and facial recognition, this technology can significantly benefit users with mobility challenges and create a more intuitive computing environment.